COSC 328: Linux Networking

Practice Midterm Key 2026

Name: _____ Student Number: _____

This midterm has three sections: Multiple choice, short answer, and long answer. For the multiple-choice section, circle the correct answer. For the short answer section, please provide your answer in short paragraphs, point form, or a query, as appropriate for the question. For long answer, provide answer as specified.

Multiple Choice: _____ / 17

Short Answer: _____ / 19

Long Answer: _____ / 15

Bonus Mark: _____ / 1

Total Mark: _____ / 51

- **Ensure your name and student number are on the midterm!**
- Answer what you know first, then come back and work on the questions you aren't sure on.
- If you have a question, feel free to raise your hand and ask.
- You may write in pen or pencil.

Use of electronic devices such as mobile phones or laptop computers is prohibited during the midterm and being caught using them will result in a mark of zero and a referral to the faculty administration.

- Good Luck! -

Section A: Multiple Choice (17 Marks Total)
Circle the correct answer

1. If someone wanted to engage in ***banner grabbing***, they would need to perform a _____ port scan.
    a. Syn
    b. UDP
    c. Full handshake
    d. Aggressive

2. When configuring a ***Fail2Ban*** jail, the ***findtime*** parameter sets…
    a. The number of allowed violations
    b. The length of the window used to search for violations
    c. The amount of time to ban violators for
    d. None of the above

3. ***Ports*** are associated with which level of the ***OSI*** model?
    a. Level 5
    b. Level 4
    c. Level 3
    d. Level 2

4. Which of the following is the correct way to allow traffic inbound via port ***1337*** using the ***UDP*** protocol using ***ufw***?
    a. sudo ufw allow 1337/udp
    b. sudo ufw -INPUT 1337/udp
    c. sudo ufw -I -t 1337
    d. sudo ufw allow -t 1337

5. If we wanted to find out information about the ***physical network interface*** of a Linux server, which of the following commands is most relevant?
    a. arp
    b. ifconfig
    c. tcpdump
    d. lshw

6. Which of the following events is ***not*** part of a ***UDP server*** socket ***lifecycle***?
    a. bind()
    b. recvfrom()
    c. sendto()
    d. accept()

7. A ***raw*** socket directly connects the ***application layer*** to the…
    a. Transport layer
    b. Data Link layer
    c. Network layer
    d. Session layer

8. Which of the following ***SMTP*** service types would best describe an ***end-user email client*** such as ***Outlook*** or ***Thunderbird***?
    a. MDA
    b. MUA
    c. MTA
    d. MSA

9. Which of the following ***protocols*** exists at the ***presentation*** layer?
    a. ARP
    b. TLS
    c. NTP
    d. TCP

10. If our firewall implemented ***stateful filtering***, this would mean that…
    a. Our firewall will automatically block malicious traffic
    b. Our firewall will prevent port scanning attempts
    c. Our firewall will allow traffic through based on a regular expression
    d. Our firewall will allow traffic through if we initiate the connection

11. Which of the following sentences best describes the ***traceroute*** function?
    a. Measures the round-trip time between us and a destination
    b. Returns all iptables filter rules applied to a given port and interface
    c. Identifies servers and routers between us and a destination
    d. Identifies the true source of a UDP datagram

12. When using the ***iptables*** command to set new filter rules, these rules are ***persistent*** and will remain in place after server reboots.
    a. True
    b. False

13. If we ran the ***ss -tulpn*** in our terminal and we saw a ***socket*** with a ***state*** listed as ***UNCONN***, this implies that the socket…
    a. Is a UDP socket
    b. Is a TCP socket
    c. Has been closed and is no longer in use
    d. Is not currently bound to a port

14. An ***OnCalendar*** parameter in a ***Systemd*** timer configuration looks like the following: ****-*-* 13:00:00***. What does this mean?
    a. The timer will be executed once per day at 1PM
    b. The timer will be executed once per week at 1PM
    c. The timer will be executed once per month at 1PM
    d. The timer will be executed once per year at 1PM

15. What type of ***DNS*** record is used to look up a recipient's ***MTA***?
    a. A
    b. AAAA
    c. MX
    d. TXT

16. Which of the following is ***not true*** of ***port scanning***?
    a. TCP port scanning is typically more reliable than UDP scanning
    b. Port scanning serves no purpose outside of malicious use
    c. It is not possible to prevent port scanning from occurring
    d. All blocked port access attempts are logged by UFW

17. When we receive a ***local*** email from within our Linux server, it is stored in the _____ before reading, and in the _____ after being read.
    a. Mail spool, user mailbox
    b. User mailbox, mail spool
    c. MDA, MUA
    d. MUA, MDA

Section B: Short Answer (19 Marks Total)
Answer the questions in short sentences, point form, or queries, as specified

18. Explain what ***swap space*** is in a Linux server context and ***why*** it is used. (***2 Marks***)

    It dedicates part of the storage memory (i.e. disk) to use as a space for swapping data in and out of RAM. +1

    It is used because it gives us a cheaper (albeit much slower) alternative to more RAM in our servers and helps prevent Out-of-Memory errors. +1

19. Name at least ***two*** user-space components of the ***Netfilter*** framework and describe what they do. (***2 Marks***)

    Any two of: ebtables, arptables, iptables, ip6tables

    - ***Ebtables – handles rules for filtering ethernet bridge traffic***
    - ***Arptables – handles rules for filtering ARP traffic***
    - ***Ip/ip6tables – handles rules for filtering IPv4/IPv6 traffic***

    +0.5 each for naming it and +0.5 each for short description

20. What is the difference between a regular ***socket*** and a ***Unix domain*** socket, and when is it appropriate to use a Unix domain socket? (***2 Marks***)

    A Unix Domain socket is a local socket – it skips all the networking layers that a normal socket goes through. +1

    It is appropriate for routing traffic between two applications running on the same host (i.e. as a local socket between two applications) +1

21. Provide an example of a ***valid nmap command*** to scan ***www.google.com***. (***1 Mark***)

nmap www.google.com +1

22. When delivering ***email***, when are the ***IMAP*** and ***POP3*** protocols used? Be specific about where in the email process this protocol is used. (***2 Marks***)

The IMAP and POP3 protocols are used for the last leg of mail delivery – between the MDA (the user's online mailbox where mail waits to be delivered to the user's client) and the MUA (the user's actual mail client like Outlook).

+1 mark each for correctly identify the servers that are communicating with these protocols (MDA and MUA)

23. Briefly describe at least ***three*** possible steps you could take to harden the security of a Linux server, based on what we've covered in this class. (***3 Marks***)

Any three of the following (and possibly more I haven't listed):

- ***Changing SSH to public key authentication***
- ***Changing SSH port***
- ***Disabling the root user account login***
- ***Using SSH-audit to verify the SSH configuration***
- ***Setting up UFW as a firewall to block unwanted traffic***
- ***Setting up Fail2Ban to ban attempted SSH logins***
- ***Setting up Fail2Ban to ban attempted port scans***
- ***Etc…***

+1 Mark each

24. Provide at least two different types of ***filters*** that can be used with the ***Fail2Ban*** package. (***2 Marks***)

Examples from the lecture slides, any two, +1 mark each:

- ***SSH (filtering SSH logs for failed logins)***
- ***UFW (filtering UFW logs for failed port scans)***
- ***Exploit detection (filtering Apache/Nginx for SQLi, code injection, etc.***
- ***Bad bots (filtering Apache/Nginx based on known bad bot user agents)***

25. Name any ***three*** of the tables within ***iptables***. (***3 Marks***)

Any three of the following options, +1 mark each:

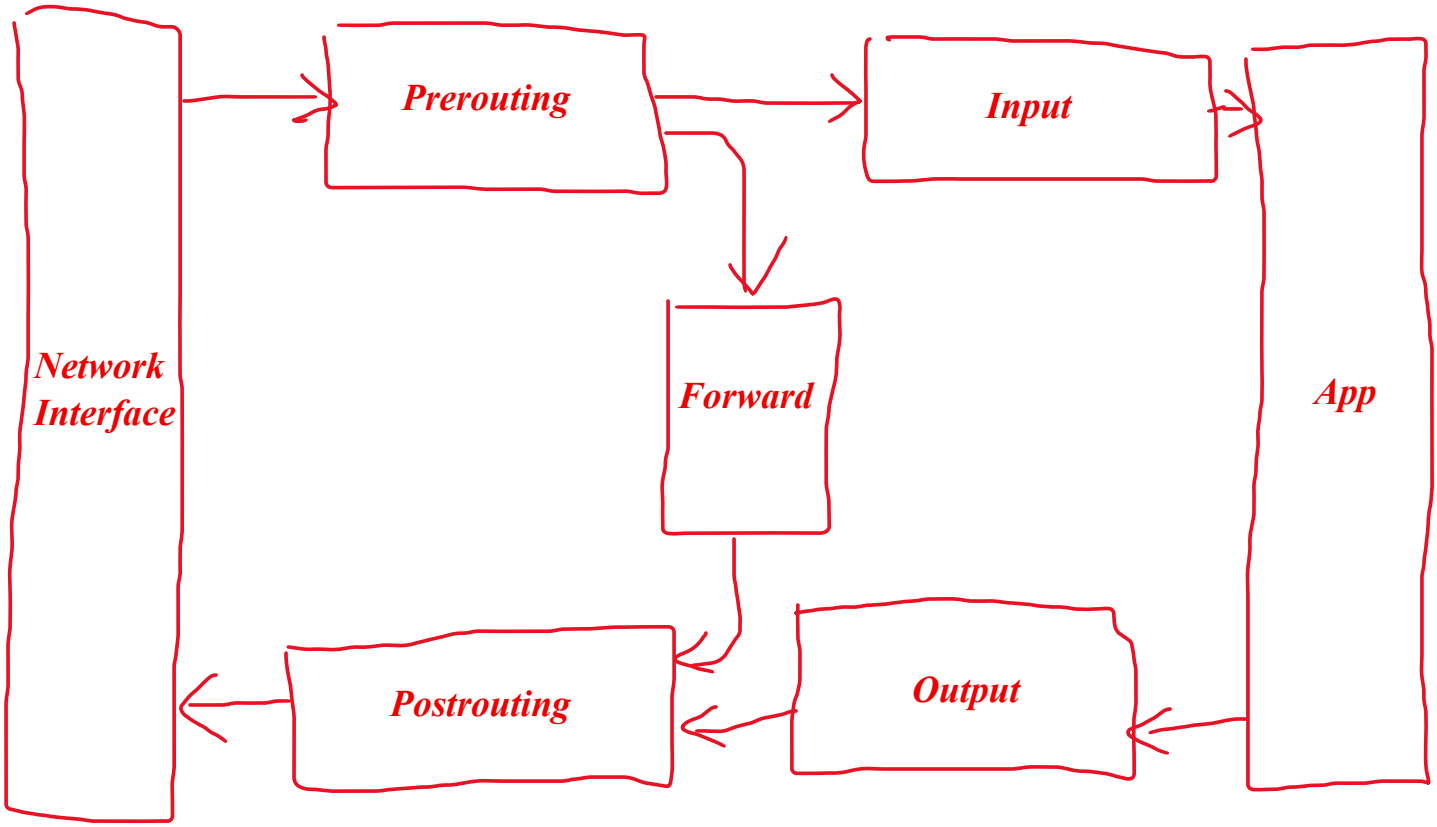
- ***Raw***
- ***Mangle***
- ***NAT***
- ***Filter***
- ***Security***

26. Briefly explain what kind of ***issues*** or ***errors*** might be discovered by the ***ssh-audit*** utility. (***2 Marks***)

SSH-audit will detect issues with the SSH configuration (+1 mark), such as issues with obsolete/insecure encryption/host key/key exchange/etc. algorithms being used (+1 mark).

Section C: Long Answer (15 Marks Total)
Answer the questions with formulas, solutions, or paragraphs as required.

27. Draw a ***simplified diagram*** showing the ***five major blocks*** of the ***iptables*** packet filter system, including the correct label for each block as well as arrows indicating connections. (***5 Marks***)



+1 mark each for prerouting, input, forward, output, and postrouting; but they must be in the correct spot with arrows in the correct direction or marks will be deducted.

28. Consider the following requirements for blocking malicious traffic:
    - We want to protect an ***SSH*** socket on ***port 6767***
    - We want to ban any user who fails more than ***10 times*** in ***5 minutes***
    - We want to ban users for ***60 minutes***
    - The ***logfile*** for the SSH service is at ***/var/log/ssh_auth.log***

    Write an appropriate ***Fail2Ban jail file*** to meet the above requirements. (***5 Marks***)

```
[ssh]
enabled = true
port = 6767
filter = sshd
logpath = /var/log/ssh_auth.log
maxretry = 10
findtime = 5m
bantime = 60m
```

+ 0.5 mark per required line (including [ssh] name)
+ 1 mark for correct syntax throughout

29. Refer to the image below for questions A through Z below.

```
eth0: flags=4163<UP, BROADCAST, RUNNING, MULTICAST> mtu 1500
    inet 10.137.0.7 netmask 255.255.0.0 broadcast 10.137.255.255
    inet6 fe80::3:deff:fead: beef prefixlen 64 scopeid 0x20<link>
    ether 02:03:de: ad:be:ef txqueuelen 1000 (Ethernet)
```

a. What ***command-line utility*** was used to generate this data? (***1 Mark***)

ifconfig

b. What is the ***physical address*** of this interface? (***1 Mark***)

02:03:de:ad:be:ef

c. What is the ***network address*** of this interface? (***1 Mark***)

Could use

10.137.0.7 (IPv4)

or

fe80::4:deff:fead:beef (IPv6)

d. Assume that this came from an actual ***VPS*** server like those we use in the labs. What kind of ***network interface*** hardware is this most likely? Justify your answer. (***2 Marks***)

If this was a VPS (like our Ubuntu servers), this would most likely be a virtual network interface, as our server isn't its own dedicated machine – it's a virtual server, sharing the hardware with hundreds of other users. So probably a virtual ethernet interface.

+1 mark for saying it'd be a virtual interface, +1 mark for justification (because it's not a real machine)

Bonus Mark: What does ***POP3*** stand for?

Post Office Protocol, version 3

- End of Practice Midterm -